Flexibility Estimation and Control of Thermostatically Controlled Loads With Lock Time for Regulation Service

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***Abstract*—A virtual battery model is a simple and general method to quantify aggregate flexibility from thermostatically controlled loads (TCLs), enabling grid operators to effectively coordinate a large number of flexible building loads with supply-side resources in power systems. Lockout controls are designed to avoid wear and tear resulting from short-cycling of hardware. The lock on/off time could significantly affect aggregate flexibility from TCLs to provide ancillary services and may even fail control algorithms designed without considering the lock time constraints. This paper focuses on flexibility estimation and control design for TCLs with lock time constraints to provide frequency regulation service. We first investigate the potential impacts of lock time on TCLs' aggregate flexibility and control performance. Both control-dependent and control-independent power bounds are then derived, based on previous TCL switching operations and regulation signals, respectively. While the control-dependent method provides aggregate flexibility for a given control method, the control-independent method calculates the theoretical maximum of power bounds. Finally, two control algorithms are proposed to better distribute flexibility over time and thereby improve signal tracking performance. The proposed methods are illustrated and validated through simulations.**

***Index Terms*—Aggregate flexibility, control, frequency regulation, lock time, thermostatically controlled loads, virtual battery.**

NOMENCLATURE

α	Self-discharging rate of virtual battery $(1/h)$
C_i	Thermal capacitance of TCL i (kWh/°C)
COP_i	Coefficient of performance of TCL i
Δ_i	Temperature deadband of TCL i (°C)
Δt	Time step size (h)
E	Energy state of virtual battery (kWh)
$\overline{E}_{\text{th}}$	Energy upper threshold (kWh)
$\underline{E}_{\text{th}}$	Energy lower threshold (kWh)
l_i^{off}	Number of time steps in lock-off mode of TCL i
l_i^{on}	Number of time steps in lock-on mode of TCL i
N	Number of TCLs
p	Virtual battery charging/discharging power (kW)
P_i	Rated power of TCL i (kW)
P^*	Desired power consumption (kW)
P_{base}	TCL baseline power consumption (kW)
$P_{\text{lb},c}$	Control-dependent power lower bound (kW)
$P_{\text{lb},s}$	Control-independent power lower bound (kW)
P_{lockoff}	Cumulative power decrease in desired power profile (kW)
P_{lockon}	Cumulative power increase in desired profile (kW)
P_{min}	Total power consumption of TCLs that are in lock-on mode (kW)
P_{max}	Total power consumption of TCLs that are not in lock-off mode (kW)
$P_{\text{ub},c}$	Control-dependent power upper bound (kW)
$P_{\text{ub},s}$	Control-independent power upper bound (kW)
R_i	Thermal resistance of TCL i (°C/kW)
s_i	On/off state of TCL i
t_i^{off}	Cumulative time in off mode of TCL i (h)
t_i^{on}	Cumulative time in on mode of TCL i (h)
θ_i	Temperature of TCL i (°C)
θ_o	Outdoor temperature (°C)
$\theta_{r,i}$	Temperature setpoint of TCL i (°C)
w_i	Thermal disturbance of TCL i (kW).

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I. INTRODUCTION

WITH a growing emphasis on energy security and environmental protection, renewable energy has been developing rapidly around the world. The increasing penetration of renewable generation imposes significant challenges on power system operation due to its natural uncertainty and variability. Additional reserves for frequency regulation and load following are needed to maintain reliable grid operation. These reserves can be provided by supply-side resources such as traditional fossil fuel generators. The disadvantages of this strategy include increased carbon emission, reduced generation efficiency, and increased cost. Exploiting flexibility from demand-side resources to provide ancillary services represents an innovative solution for the power grid.

Among all loads, thermostatically controlled loads (TCLs) consume about 20% of the electricity in the United States [1]. When properly controlled, TCLs with an inherent ability to store heat in the thermal mass can vary their power consumption with little impact on customer convenience or comfort.

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TCLs represent a significant and largely untapped resource for providing ancillary services in power systems. There is abundant literature on controlling TCLs for regulation and load following services. For example, a high-level supervisory control strategy is proposed in [2] to direct heating, ventilation, and air-conditioning (HVAC) systems for frequency regulation with strengthened interactions and depressed counteractions. In [3], a state bin is used to model a population of TCLs, and a model predictive control (MPC) scheme is designed to optimally control TCLs for frequency regulation. Adaptive control approaches are developed in [4] to enable TCLs to better track signals for load following in the presence of heat disturbance. A hierarchical control framework is proposed in [5] for integrated coordination between TCLs and other distributed energy resources to provide load-following service. In [6], controllers are designed for domestic refrigerators to provide primary and secondary frequency response. A distributed control is proposed in [7] to provide generation following in the real-time energy market. In [8], the authors propose two control strategies to enable HVAC systems to provide frequency regulation services: an MPC for variable air volume HVAC systems and a rule-based control for aggregation of on/off HVAC systems.

While these methods can be used to schedule and control a small group of TCLs to provide ancillary services, integrating a large population of TCLs into system-level operation and control is still challenging. Models and methods for quantifying aggregate flexibility are critical for grid operators to effectively coordinate TCLs with generators and energy storage systems. During the past few years, several studies have been devoted to quantification and characterization of aggregate flexibility from TCLs. In [9], an analytical approach is developed to characterize and control the reserve capacity of TCLs. A demand-side response model is proposed for TCLs to enable optimal scheduling of power and energy consumption and provide multiple ancillary services in [10]. A *virtual battery* (VB) model has also been proposed to quantify flexibility from building loads with diversified dynamics, constraints, and characteristics. The VB is a scalar linear system that resembles simplified battery dynamics parameterized by charging/discharging power limits, energy limits, and self-discharging rate. Analytical methods are presented in [11] to estimate aggregate flexibility as VB from homogeneous and heterogenous on/off TCL systems. In [12], a geometric approach is proposed to improve the estimation accuracy of aggregate flexibility from heterogeneous TCLs. To estimate VB parameters for more complex commercial HVAC systems, a simulation-based method using detailed building models is proposed in [13], and an optimization-based method using simplified building models is proposed in [14]. Once VB parameters are identified, existing scheduling and coordination methods such as the ones in [15]–[17], can be adapted for scheduling and coordination at the system level. The VB model is also useful for quantifying regional flexibility and understanding potential economic benefits from building loads [18]. Nevertheless, in these VB-related works, one of the most important practical constraints, the lock time constraints, are not fully addressed. In practice, in order to avoid wear and tear resulting from short-cycling of TCLs, lockout controls are designed to prevent devices (e.g., compressors and supply fans in HVAC) from being kicked on/off until they meet the minimum time duration between two consecutive on/off switches. This lock time is modeled when evaluating TCLs control to deliver ancillary services in [19] and [20]. When it comes to the flexibility estimation, [21] develops new power and energy bounds for VB models but only considers large time steps, which are assumed to be considerably longer than the lock time. The authors in [22] derive ramping rate constraints for VB models and adopt the priority-stack based (PSB) control scheme to coordinate TCLs to follow the desired power consumption. Nevertheless, it is not clear whether existing control methods, e.g., the PSB control, can fully utilize the VB flexibility considering lock time constraints.

Ignoring the unavailability of TCLs in lock modes could result in inappropriate on/off control of devices and therefore degrade performance of regulation services. More importantly, when a control algorithm fails to follow a regulation signal from the grid, it is not clear whether the failure is because the desired response is beyond the aggregate flexibility of TCLs, or the control method cannot properly utilize the flexibility from TCLs. To address these issues, this paper studies the impact of lock time on aggregate flexibility from TCLs and incorporates lock time constraints into flexibility estimation and control design. The contributions of this paper are twofold.

- Control-dependent and control-independent aggregate flexibility are derived considering lock time constraints. The control-dependent flexibility is determined based on previous on/off operation of TCLs for a given control algorithm. The control-independent flexibility only depends on a regulation signal profile and represents the theoretical maximum of power ranges.
- Two control algorithms are proposed to better exploit aggregate flexibility from TCLs with lock time constraints. Compared with the existing PSB control, the proposed control algorithms are able to better distribute flexibility over time based on the need of regulation service.

The rest of this paper is organized as follows. The VB model is briefly reviewed in Section II. Section III studies the impacts of lock time on aggregate flexibility and TCL control for frequency regulation through a case study. In Section IV, control-independent and control-dependent power bounds are derived considering lock time constraints. Two control methods are proposed in Section V to mitigate the negative impacts of the lock time in providing regulation service. The proposed flexibility estimation and control are illustrated and compared with existing methods through case studies in Section VI. Conclusions and future work are offered in Section VII.

II. Virtual Battery Model

This section presents the VB model used to characterize aggregate flexibility of TCLs, taking an air conditioner (AC) in cooling mode as an example. The indoor air temperature

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TABLE I
VIRTUAL BATTERY PARAMETERS

Energy state dynamics	$\dot{E}(t) = -\alpha E(t) + p(t)$
Energy upper bound	$\sum_{i=1}^{N}(1 + \lvert 1 - \frac{1}{R_i C_i}\frac{1}{\alpha}\rvert)\frac{C_i \Delta_i}{COP_i}$
Energy lower bound	negative of energy upper bound
Power lower bound	$\sum_{i=1}^{N} P_i - P_{\text{base}}$
Power upper bound	$-P_{\text{base}}$

dynamics of an AC with index i can be represented by a first-order ordinary differential equation:

$$C_i \frac{d\theta_i(t)}{dt} = \frac{\theta_o(t) - \theta_i(t)}{R_i} - s_i(t) P_i COP_i + w_i(t), \quad (1)$$

where θ_i is the indoor air temperature, θ_o is the outdoor temperature, C_i is thermal capacitance, R_i is the thermal resistance, s_i is the AC on/off state that is unity when AC is on and zero when AC is off, P_i is the rated power, COP_i is the coefficient of performance, and w_i is the external disturbance. Each AC has a temperature setpoint $\theta_{r,i}$ with a hysteretic ON/OFF local control within a temperature band $[\theta_{r,i} - \Delta_i, \theta_{r,i} + \Delta_i]$. The operating state s_i evolves as a discrete function of air temperature θ_i. It was shown in [11] that for a large population of ACs, their aggregate behavior with the hybrid model can be approximated by using the continuous model. The baseline power consumption from these ACs can be approximated as

$$P_{\text{base}}(t) = \sum_{i=1}^{N} \frac{\theta_o(t) - \theta_{r,i}}{COP_i R_i}, \quad (2)$$

where N is the number of ACs. A VB model can be constructed to characterize the aggregate flexibility from a group of ACs. It defines a set of power profiles $p(t)$ that satisfy constraints listed in Table I [11], where $\alpha = N/(\sum_{i=1}^{N} R_i C_i)$ is the self-discharge rate, capturing varying changes in thermal losses with increased or decreased temperatures from setpoints.

III. IMPACTS OF LOCK TIME

Lockout controls are designed to avoid wear and tear resulting from short-cycling of TCLs components, such as compressors and supply fans. A TCL is in lock-on (or lock-off) mode when it is switched "on" (or "off") until the minimum duration time is met. This section evaluates the impacts of lock time on the VB model and the PSB control through a case study. Only lock-off time constraints are considered in this example, but lock-on time constraints can be studied similarly. Historical outdoor air temperature in Columbus, Ohio is used, as plotted in Fig. 1. A group of homogeneous ACs are considered with typical parameters adopted from [23], as listed in Table II. For these homogeneous ACs, the VB model accurately represents their aggregate flexibility when there is no lock time. The VB parameters are calculated according to Table I. A regD signal [24] in PJM Interconnection is scaled to generate the desired power deviation that is within the VB flexibility for regulation service. The desired deviation and

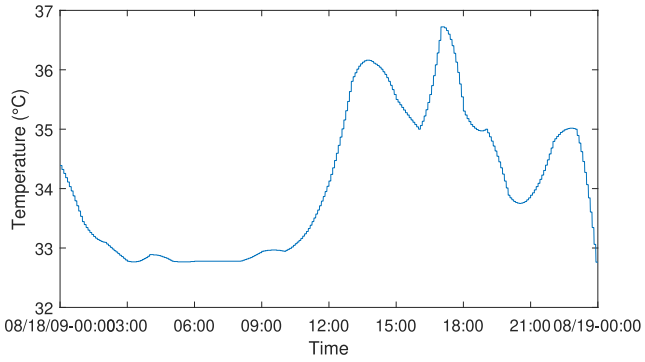


Fig. 1. Outdoor air temperature.

TABLE II
AC PARAMETERS

Parameter	Value
Thermal capacitance	2 kWh/°C
Thermal resistance	2 °C/kW
Rated power	5.6 kW
Coefficient of performance	2.50
Temperature setpoint	22.5°C
Temperature deadband	0.5°C
Lock-off time	2 minutes
Number of ACs	1000

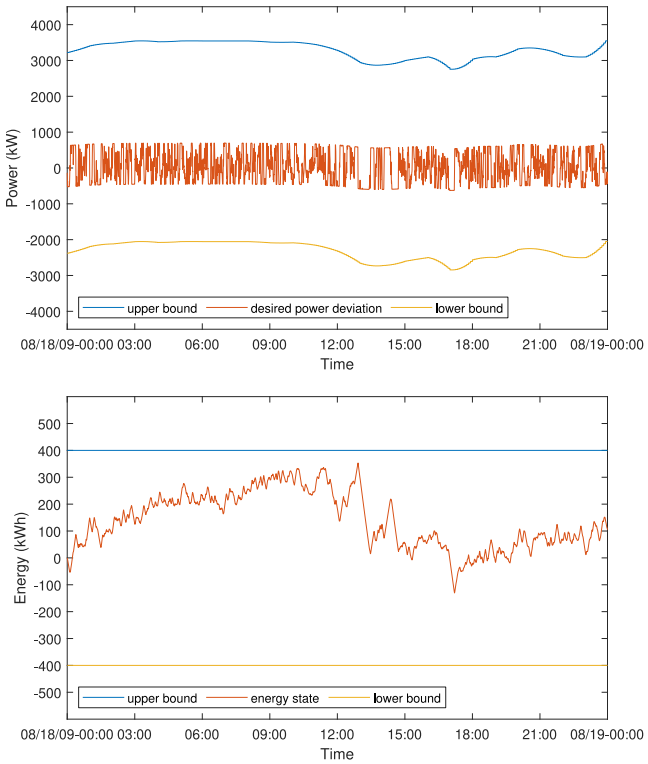


Fig. 2. Regulation signal vs VB power bounds (top); corresponding energy state vs VB energy bounds (bottom).

VB bounds are plotted in Fig. 2. As can be seen, the desired deviation is within both the power and energy limits without considering lock time constraints.

The PSB algorithm is used to control ACs with and without lock-off time to provide desired power consumption for

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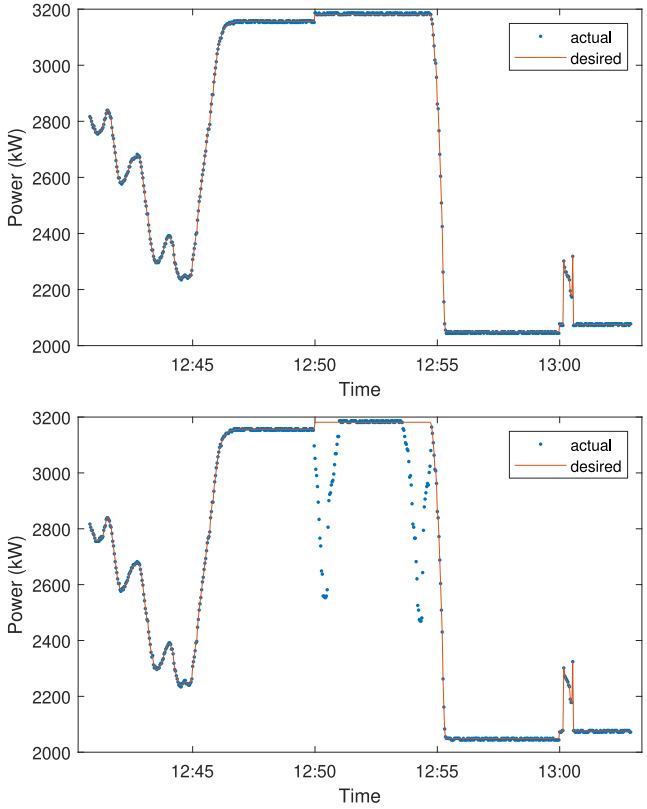


Fig. 3. Signal-following performance of the PSB control without lock-off time (top) and with lock-off time (bottom).

regulation service while maintaining the indoor air temperature within the deadband. When there is no lock-off time, the desired power consumption can be followed well. On the other hand, with a two-minute lock-off time, the PSB algorithm fails to control these ACs to follow the desired power consumption in some time periods. The results for a sampled 25-minute period are shown in Fig. 3. It is found that many ACs are in lock-off mode between 12:50 and 12:55 with lock-off time constraints and there are not enough ACs to turn on to provide the desired power consumption.

This simple yet representative case study shows that the PSB algorithm fails to control TCLs with lock time to respond to a regulation signal. Nevertheless, it is not clear whether the failure is because the desired response is beyond the aggregate flexibility of TCLs with lock time, or the control method cannot properly utilize the flexibility from TCLs. Theoretical work is required to fully understand the impacts of lock time on aggregate flexibility and the effectiveness of control algorithms for TCLs to provide regulation service.

IV. Proposed Flexibility Estimation Methods

With lock time constraints, some TCLs temporarily cannot be turned on or off. Therefore, the aggregate power flexibility bounds need to be adjusted accordingly. In this section, theoretical results are presented on flexibility estimation for TCLs with lock time constraints. Two sets of power bounds are derived. The control-dependent power bounds are determined based on previous on/off operation of TCLs for a given control algorithm. The control-independent power bounds only depend on a regulation signal profile and represent the theoretical maximum of power ranges. Please note that the derived power bounds can be applied to both homogeneous and heterogeneous populations of TCLs.

A. Control-Dependent Power Bounds

Without lock time constraints, all TCLs can be turned on or off at any time to increase or decrease the total power consumption. With lock time constraints, TCLs that are in lock modes cannot be turned on or off and therefore become temporarily unavailable for adjusting total power consumption. Lock time constraints thus reduce the power flexibility of TCLs.

For a group of TCLs, the maximum power consumption is the sum of the rated power of TCLs that are not in lock-off mode, as expressed in (3).

$$P_{\max}(t) = \sum_{\{i: t_i^{\text{off}}(t) \geq l_i^{\text{off}} \Delta t\}} P_i, \tag{3}$$

where t is the time step index, Δt is time step size, $t_i^{\text{off}}(t)$ is the time of the i–th TCL in off mode by time t, and $l_i^{\text{off}} \Delta t$ is the lock-off time of the i–th TCL. The power upper bound in the VB model is the difference between the maximum power consumption and the baseline, i.e.,

$$P_{\text{ub,c}}(t) = P_{\max}(t) - P_{\text{base}}(t). \tag{4}$$

The minimum power consumption is the sum of the rated power of TCLs that are in lock-on mode, as expressed in (5).

$$P_{\min}(t) = \sum_{\{i: t_i^{\text{on}}(t) \leq l_i^{\text{on}} \Delta t\}} P_i, \tag{5}$$

where $t_i^{\text{on}}(t)$ is the time of the i–th TCL in on mode by time t, and $l_i^{\text{on}} \Delta t$ is the lock-on time of the i–th TCL. The power lower bound in the VB model is the difference between the baseline and the minimum power consumption, i.e.,

$$P_{\text{lb,c}}(t) = P_{\min}(t) - P_{\text{base}}(t). \tag{6}$$

The power upper bound in (4) and lower bound in (6) are determined based on the operation and status of the TCL population, which depend on control methods. They are thus called control-dependent power bounds. For a given control algorithm, at any time step t, when the desired power deviation exceeds these bounds, TCLs cannot be controlled to follow the signal without violating thermal comfort requirements.

B. Control-Independent Power Bounds

With lock time constraints, ACs could be turned off and stay in the lock-off mode for i) maintaining the temperature within the deadband, or ii) following regulation signals. The control-dependent power bounds capture both. Different control algorithms may lead to different control-dependent power bounds. When the desired power deviation exceeds the power bounds of a control algorithm, it is not clear whether there exist other control algorithms that can follow the signal. Therefore, a method is proposed next to quantify the

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theoretical maximum of power bounds regardless of control methods. The control-independent power bounds are derived based on the minimum power associated with switching operations required to follow a regulation signal, as given in Theorem 1.

Theorem 1: For a group of TCLs with dynamics described in (1) and parameters C_i, R_i, P_i, COP_i, $\theta_{r,i}$, Δ_i, $i = 1, 2, \ldots, N$, with a lock-off time equal to $l_{off}\Delta t$ and a lock-on time equal to $l_{on}\Delta t$, for a given desired power consumption $P^*(t)$, the VB power upper bound is

$$P_{ub,s}(t) = \sum_{i=1}^{N} P_i - P_{lockoff}(t) - P_{base}(t), \tag{7}$$

where $P_{lockoff}(t) = \sum_{j=-l_{off}}^{-1} \max\{0, P^*(t+(j-1)\Delta t) - P^*(t+j\Delta t)\}$. The VB power lower bound is

$$P_{lb,s}(t) = P_{lockon}(t) - P_{base}(t), \tag{8}$$

where $P_{lockon}(t) = \sum_{j=-l_{on}}^{-1} \max\{0, P^*(t+j\Delta t) - P^*(t+(j-1)\Delta t)\}$.

Proof: We first derive the VB power upper bound, which is the difference between the maximum power consumption and the baseline. Without lock time constraints, all TCLs can be temporarily turned on, and therefore the maximum power consumption is the total rated power of all TCLs, which is $\sum_{i=1}^{N} P_i$. With lock constraints, some TCLs that are in lock-off mode cannot be turned on, and therefore their rated power needs to be removed from the maximum power consumption. To estimate the VB power upper bound, we only need to calculate the total rated power from TCLs that are in lock-off mode.

At any time t, TCLs in lock-off mode are those being turned off from time $t - l_{\text{off}}\Delta t$ to t. At any time $\tau \in [t - l_{\text{off}}\Delta t,\ t-1]$, when $P^*(\tau) \geq P^*(\tau-1)$, there is no need to turn off any TCLs for the purpose of providing regulation service. Therefore, providing regulation service does not cause any TCLs to be in lock-off mode. On the other hand, when $P^*(\tau) < P^*(\tau-1)$, a number of TCLs have to be turned off to reduce power consumption at time τ. Therefore, providing regulation service causes a number of TCLs to be in lock-off mode and at least $P^*(\tau-1) - P^*(\tau)$ power is removed from the maximum power consumption from τ to $\tau + l_{\text{off}}\Delta t$. Thus, the total rated power from TCLs that are in locked-off mode is $\sum_{j=-l_{\text{off}}}^{-1} \max\{0, P^*((j-1)\Delta t) - P^*(j\Delta t)\}$, and the VB power upper bound in (7) is obtained. The VB power lower bound is affected by lock-on time constraints and can be derived similarly. ∎

The power bounds in Theorem 1 only take into account TCL switching operation for the purpose of providing regulation service and do not consider the switching operation to maintain the temperature within the deadband. Therefore, the obtained power bounds represent a theoretical maximum of VB flexibility.

V. Proposed Control Algorithms

To mitigate the negative impacts of lock time constraints on providing regulation service, we propose a new control algorithm named modified PSB (mPSB) to better distribute flexibility over time based on the regulation need and thereby improve signal tracking performance. In addition, a hybrid algorithm is developed by combining the mPSB and PSB control to lower switching frequency.

Algorithm 1 Modified PSB Control

Input: Desired power consumption P^*

1: Turn on TCLs that are in lock-on mode or with temperatures above the upper bounds. The total rated power of these TCLs is denoted as $P_{\text{must-on}}$.
2: Turn off TCLs that are in lock-off mode or with temperatures below the lower bounds.
3: Randomly select a portion of TCLs from the rest and sort the selected TCLs by temperature in descending order. Denote the sorted TCLs as TCL_{p}. The total rated power of these TCLs is denoted as P_{p}.
4: Group the remaining TCLs based on their on/off status at previous time step and sort each group by temperature in descending order. Denote the on-group TCLs as TCL_{on} and the total rated power as P_{on}. Denote the off-group TCLs as TCL_{off} and the total rated power as P_{off}.
5: **switch** (P^*)
6: **case** $P_{\text{must-on}} \leq P^* < P_{\text{must-on}} + P_{\text{on}}$**:**
7: Select the first several TCLs in TCL_{on} until the total power consumption is equal to $P^* - P_{\text{must-on}}$.
8: **case** $P_{\text{must-on}} + P_{\text{on}} \leq P^* < P_{\text{must-on}} + P_{\text{on}} + P_{\text{p}}$**:**
9: Select all TCLs in TCL_{on}; select the first several TCLs in TCL_{p} until the total power consumption is equal $P^* - P_{\text{must-on}} - P_{\text{on}}$.
10: **case** $P_{\text{must-on}} + P_{\text{on}} + P_{\text{p}} \leq P^* < P_{\text{must-on}} + P_{\text{on}} + P_{\text{p}} + P_{\text{off}}$**:**
11: Select all TCLs in TCL_{on} and TCL_{p}; select the first several TCLs in TCL_{off} until the total power consumption is equal $P^* - P_{\text{must-on}} - P_{\text{on}} - P_{\text{p}}$.
12: **case** $P_{\text{must-on}} + P_{\text{on}} + P_{\text{p}} + P_{\text{off}} \leq P^*$**:**
13: Select all TCLs in TCL_{on}, TCL_{p}, and TCL_{off}.
14: **end switch**
15: Turn on the selected TCLs and turn off the remaining ones in TCL_{on}, TCL_{p}, and TCL_{off}.

With lock time constraints, ACs could be turned off and stay in the lock-off mode for maintaining the temperature within the deadband in addition to the purpose of following regulation signals. A control algorithm may unnecessarily turn off too many ACs within a short time period, resulting in poor signal-following performance. In Fig. 3, the off-tracking between 12:50 and 12:55 indicates insufficient flexibility for that time period. On the other hand, there could be redundant flexibility in other time periods. If a control method can shift some power flexibility from time periods with redundant flexibility to time periods with insufficient flexibility, it could provide better signal-following performance. With this idea in mind, the mPSB algorithm is proposed, as given in Algorithm 1 using TCLs in cooling mode as an example. The control algorithm for TCLs in heating mode is similar and therefore is omitted here to conserve space. In this method, a portion of the TCLs are randomly selected and turned on/off

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Algorithm 2 Hybrid Control

Input: Desired power consumption P^*, an energy upper threshold $\overline{E}_{\text{th}}$, and an energy lower threshold $\underline{E}_{\text{th}}$

1: **if** $E > \overline{E}_{\text{th}}$ **or** $E < \underline{E}_{\text{th}}$, where E is the energy state, **then**
2: mPSB is active
3: **else**
4: PSB is active
5: **end if**

only based on temperature, regardless of their previous on/off states. In this way, some TCLs are switched before they reach the temperature bounds so that they fall into lock mode during time periods with redundant flexibility but will become available later when there is insufficient flexibility. The remaining TCLs are divided into "on" and "off" stacks and controlled in a similar manner to that of the PSB algorithm.

Remark 1: The main computational cost of Algorithm 1 includes sorting TCLs according to temperature in Steps 3 and 4, and cumulative sum and comparisons in Steps 7, 9, and 11, with a computational complexity depending on the number of TCLs N.

- The computational complexity of sorting is $O(N \log N)$ [25].
- The computational complexity of cumulative sum is $O(N)$ [26].
- The comparison can be completed using a search algorithm to find an element in a sorted vector. The computational complexity is $O(\log N)$ [27].

The overall computational complexity of Algorithm 1 is $O(N \log N)$.

In the proposed mPSB control, some TCLs are selected to be sorted and switched purely based on their temperatures, ignoring their previous on/off status. The number of TCLs selected in Step 3 of Algorithm 1 affects how flexibility is shifted between time periods and thereby affects the signal-following performance. Furthermore, among these selected TCLs, some "on" devices could switch status with "off" devices, reducing the average switching time. The switching frequency increases with the number of selected TCLs in Step 3. Typically, only a small percentage of TCLs needs to be randomly selected at every time step, e.g., 0-5% for the 2-second frequency regulation service considered in this paper. The switching operation for the entire population is affected within a few minutes. In practice, the optimal percentage can be tuned and updated based on offline simulations using historical data.

In the PSB control, when determining which TCLs are to be on, TCLs in the "on" stack have higher priority than those in the "off" stack. Such a strategy helps to reduce switching operation. TCLs are switched more often than with the mPSB control in spite of a better signal-following performance. Increased switching frequency negatively affects the lifetime of TCL devices. To mitigate this effect, a hybrid control method is proposed by combining the mPSB and PSB, as presented in Algorithm 2. The main idea of the hybrid control is to only execute the mPSB algorithm when the flexibility is insufficient. Insufficient flexibility typically occurs when the VB energy state is close to its energy bounds. Energy thresholds are used to determine when to activate the switching between the PSB and mPSB. When the energy state is within the thresholds, PSB is used to follow the signal with a minimum number of switching operations. When the energy state exceeds the thresholds, mPSB is used to improve signal-following performance.

Remark 2: The energy thresholds in Algorithm 2 affect both signal following performance and TCL switching frequency. Ideally, we desire an upper bound (a lower bound) that is just low (high) enough to follow dispatch signals while minimizing TCL switching operation. Developing a theoretical optimization method for these thresholds is challenging and alternative heuristic methods can be used. For example, the static upper and lower bounds can be calculated using the method in [11] with average parameters of TCLs. The two thresholds can be specified as percentages of the obtained static upper and lower bounds, respectively, based on offline simulation results with historical data. Decreasing the percentage improves signal-following results, but leads to increased switching frequency, and the vice versa. In practice, TCL aggregators can select those percentages (either constant or time-varying) based on their preferences.

Please note that the algorithms proposed in this paper assume that HVAC users accept direct control from aggregators, similar to many existing load control algorithms. The willingness of customers' participation in different demand response and load control programs is important to the deployment of using building loads for grid services. It has been shown the annual revenue per unit is not very attractive if the total revenue from providing ancillary service is split evenly [23]. On the other hand, customers generally prefer a low probability large reward over a guaranteed small reward [28]. Incentivizing methods such as lottery-based [29] or coupon-based [30] incentives are promising to increase the participation of residential customers in demand response and load control programs. The design of customer incentives is beyond the scope of this paper.

VI. Case Studies

The proposed flexibility estimation and TCL control methods are illustrated and validated through the same example presented in Section III. Key results are presented and compared as follows.

A. Flexibility Estimation Results

As only lock-off time is considered in this example, the VB power upper bounds are analyzed and compared in this subsection. Control-dependent power upper bounds were generated for the PSB, mPSB, and hybrid control, using the method presented in Section IV-A. Control-independent power upper bound was generated based on the regulation signal using the method presented in Section IV-B. These results, together with the upper bound without considering lock time and

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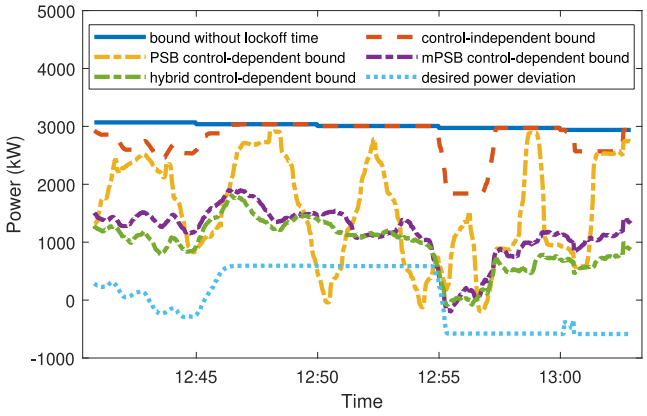


Fig. 4. Power upper bounds vs. desired power response.

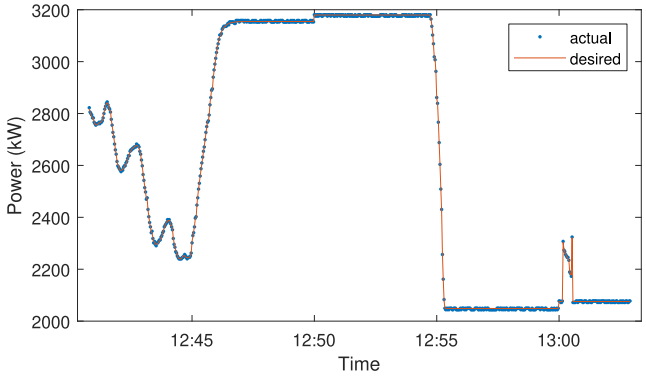


Fig. 5. Signal-following performance with the mPSB control.

desired power deviation, are plotted in Fig. 4. The following observations can be made.

- Lock time constraints reduce the aggregate flexibility. Therefore, with lock-off time, both control-dependent and control-independent upper bounds are below the one without considering lock-off time.
- The control-independent upper bound represents the theoretical maximum and it is above all control-dependent upper bounds.
- The desired power deviation exceeds the control-dependent upper bound for the PSB algorithm for a few time steps between 12:50 and 12:55. This explains why the PSB algorithm cannot control ACs to follow the desired power response in that period as shown in Section III.
- The desired power deviation is below the control-independent upper bound all the time. This means there may exist other algorithms that can control ACs to follow the regulation signal well.
- The control-dependent power upper bounds with the two proposed control algorithms are close to each other. The desired power deviation is below these two power upper bounds. This confirms that the two proposed algorithms can control ACs to follow the regulation signal. Signal-tracking results will be provided in the next subsection.
- The power upper bounds of the two proposed control algorithms are flatter than the PSB algorithm, because they better distribute the flexibility over time.
- It is also interesting to note that there is a step down in all power bounds with lock time around 12:55. This is due to the step down in the desired power deviation, which requires a number of ACs to be turned off to follow the signal. These ACs enter lock-off mode and become temporarily unavailable to be turned on, and the power upper bounds are therefore reduced.

B. TCL Control Results

Both proposed control algorithms are examined and compared with the existing PSB algorithm. In particular, 2% of TCLs are selected in Step 3 when executing the mPSB control. In the hybrid control algorithm, $\overline{E}_{\text{th}}$ is set to be 62.5% of the energy upper bound (about 250 kWh), and $\underline{E}_{\text{th}}$ is inactive because there is no lock-on time in the example. Therefore, with the hybrid algorithm, the mPSB control is only executed $E > \overline{E}_{\text{th}}$ to better track the regulation signal, while the PSB control is executed in the remaining time to help reduce the switching operation.

The signal-following performance for the mPSB algorithm is presented in Fig. 5. The signal-following profile with the hybrid algorithm is almost the same as the mPSB algorithm, and therefore is omitted. As can be seen, with the proposed algorithms, ACs with lock time constraints can be controlled to follow the regulation signal well all the time, including the time period between 12:50 and 12:55. when the PSB algorithm fails to follow the regulation signal, as shown in the second subplot in Fig. 3.

ACs switch more frequently with the mPSB control than with the PSB control. With the hybrid algorithm, when the energy state is above the energy upper threshold, ACs are operated using the mPSB control. Otherwise, ACs are operated using the PSB control. To illustrate the switching characteristics of these control algorithms, temperatures of an example AC are plotted in Fig. 6. Every turning point in temperature plot indicates a switching operation. When the temperature decreases after a turning point, it means the AC is switched on. When the temperature increases after a turning point, it means the AC is switched off. There are many more turning points in the temperature plot with the mPSB control than the PSB control. It indicates that the AC switches more frequently with the mPSB control than with the PSB control. The increasing switching frequency is because a portion of ACs are switched on/off purely based on the indoor air temperature, regardless of their previous on/off state. A slight change in temperature may lead to a switching operation, and thereby increasing switching frequency in general. The mPSB control is only active between 11:00 and 13:00 when the energy state is high, exceeding the upper threshold, as can be observed in Fig. 2. Therefore, the number of turning points and switching operations with hybrid control are about the same as the mPSB control in this period and about the same as the PSB control in the remaining time.

The average time periods between two consecutive on/off switches for all ACs are plotted for the PSB, mPSB, and hybrid

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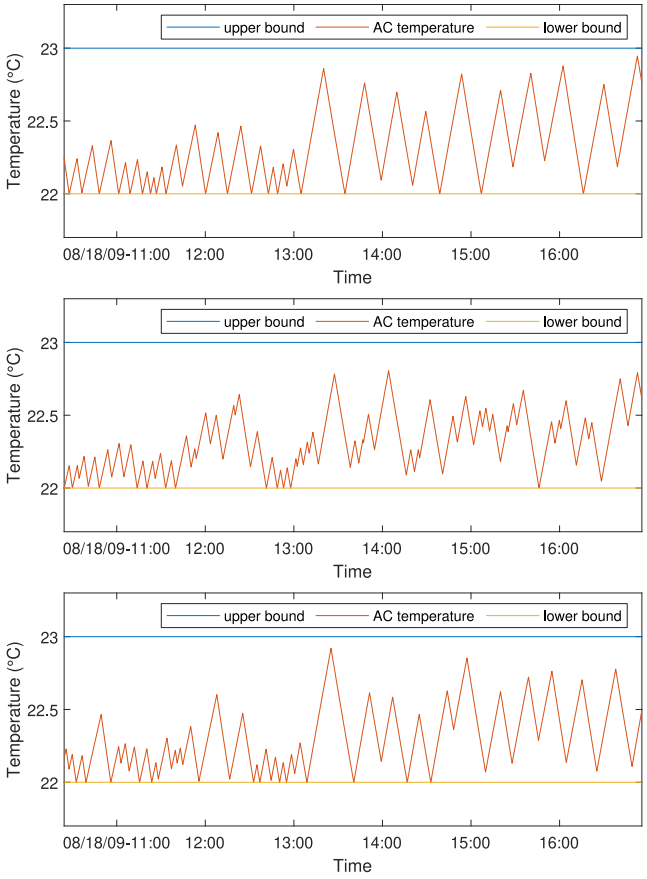


Fig. 6. Temperature with the PSB control (top), mPSB control (middle), and hybrid control (bottom).

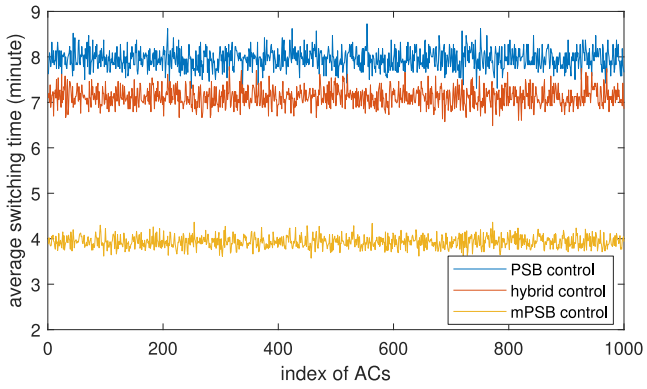


Fig. 7. Average time between two consecutive on/off switches with the PSB, mPSB, and hybrid control.

control in Fig. 7. As can be seen, the average switching durations for all ACs are at the same level for all three algorithms. This means these control algorithms fairly utilize the flexibility of ACs. In addition, the switching durations for all three algorithms are longer than the two-minute lock-off time. Moreover, compared with the PSB algorithm, the mPSB algorithm significantly reduces switching duration and increases switching frequency. The hybrid control reduces switch frequency compared to the mPSB control and provides better signal-following performance than the PSB control. The average switching

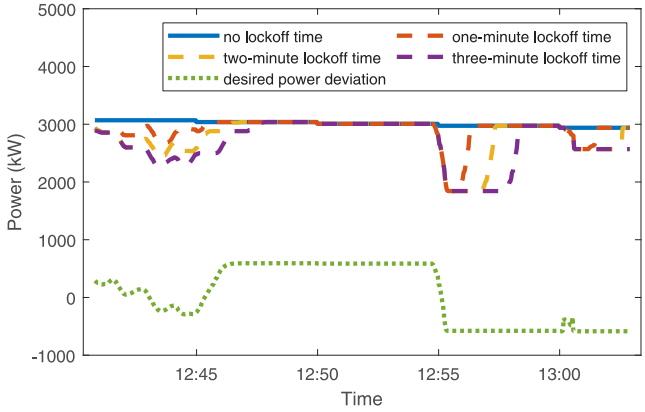


Fig. 8. Control-independent power upper bound vs. lock time.

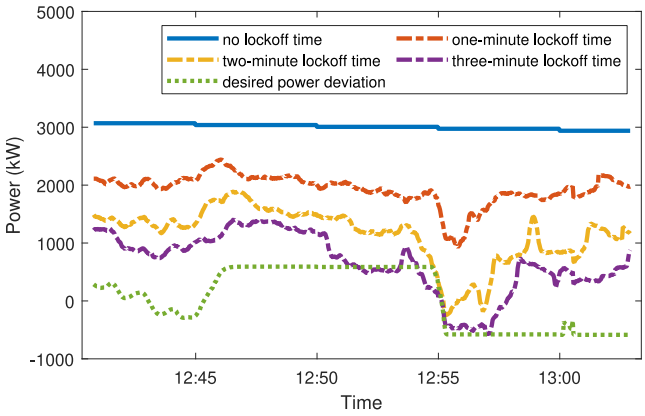


Fig. 9. The power upper bound of the mPSB control vs. lock time.

duration of the hybrid control is only about one minute less than those with the PSB control.

C. Sensitivity Analysis

Sensitivity analysis was performed to understand how the flexibility power bounds and TCLs switching time vary with lock time. The power upper bounds of the control-independent method and the mPSB control are provided in Fig. 8 and Fig. 9, respectively, with lock times equal to one, two, and three minutes. As expected, the power upper bound decreases as lock time increases, leading to less flexibility to be used. The control-independent method calculates the cumulative change in response signals within the lock time. Therefore, the changes in power upper bound purely depend on how much the response signal changes within different lock times. On the other hand, both demand response signal and the control algorithm affect the power upper bound in Fig. 9. In this example, even with the proposed mPSB control, the desired response signal cannot be exactly followed using TCLs with a lock time of three minutes. This suggests that the aggregator needs to seek a control algorithm with a better signal-following performance or to reduce the commitment in providing regulation service.

Figure 10 plots the average switching time of the TCL population for different lock times. As can be seen, the average switching time of the PSB control is insensitive to lock

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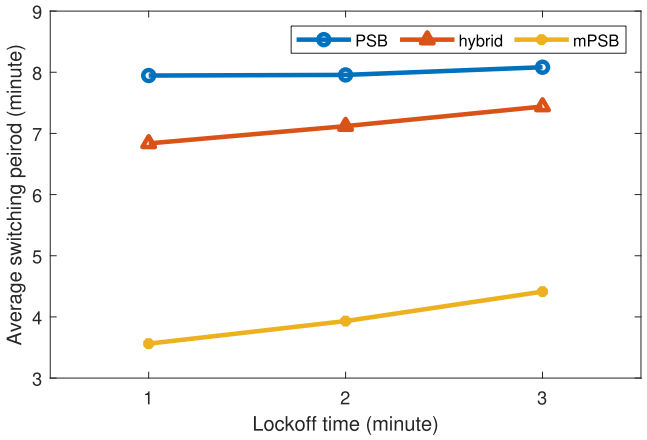


Fig. 10. Switching time vs. lock time.

time as compared the other two control algorithms. The average switching time for the mPSB and hybrid control changes within a small range and is about linear to lock time for the PJM regD signal, but the slope depends on the regulation capacity compared with the baseline power.

VII. Conclusion

This paper considered the problem of using TCLs with lock time constraints to provide frequency regulation service. We first studied the impacts of lock time on aggregate flexibility from TCLs and PSB control through a case study. Two flexibility estimation methods were then proposed considering lock time constraints, to accurately characterize power flexibility for a specific control algorithm or estimate the theoretical maximum of power ranges. In addition, two new control algorithms were designed to better exploit aggregate flexibility from TCLs with lock time constraints. It was found that the proposed control methods effectively shift flexibility from time periods with redundant flexibility to time periods with insufficient flexibility, thereby providing better signal-following performance. One area of future work is to develop an optimal control strategy to maximize revenue from regulation service considering both capacity and performance payment. Another interesting research direction is to evaluate TCLs for various grid services using high-fidelity models that more accurately capture load dynamics, heat disturbance, and various operating characteristics.

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